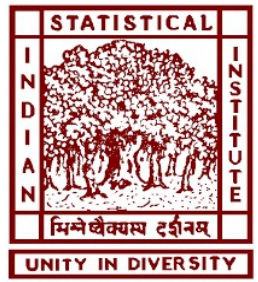
Advanced Statistical Methods II

Lecture 23: Principal Components and Principal Components Regression

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First Semester, AY 2026–27

Outline of Today's Lecture

Motivation: collinearity as a problem of directions

Principal component analysis

Principal components regression

Summary

The linear regression setup

$\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathbb{V}\mathrm{ar}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n$, with *many* or *highly correlated* predictors.

Three ways to stabilize least squares:

Strategy	Method	Idea
Shrink	ridge (Hoerl and Kennard, 1970)	pull the coefficients towards **0**
Reduce	PCR (today)	regress on a few *derived* variables
Select	LASSO, LARS	set some coefficients exactly to 0

Today:

1. Principal component analysis (PCA)
2. Principal components regression (PCR)

The Longley data: Multicollinearity

U.S. macroeconomic series, 1947–1962: `Employed` on six predictors, $n = 16$ (Longley, 1967).

- $R^2 = 0.996$: the *fit* is excellent.
- $\mathbb{C}\mathrm{orr}(\mathrm{GNP}, \mathrm{Employed}) = 0.98$, yet $\widehat{\beta}_{\mathrm{GNP}} < 0$.
- Dropping the single year 1962 changes the sign of two of the six coefficients.
- Variance inflation factors range from 4 to 1789.
- Eigenvalues of the correlation matrix of the predictors:

$$4.6, \quad 1.18, \quad 0.20, \quad 0.015, \quad 0.0026, \quad 0.00038.$$

The data cannot separate the effects of GNP, population and year: some *linear combinations* of the predictors hardly vary at all.

The spectral decomposition

Let $\boldsymbol{X}$ be centred, of rank p, with $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{V}\boldsymbol{D}^2\boldsymbol{V}^\top = \sum_{j=1}^{p} d_j^2 \boldsymbol{v}_j \boldsymbol{v}_j^\top$, $d_1 \geq \cdots \geq d_p > 0$. Then

$$\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = \sigma^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \sigma^2 \sum_{j=1}^{p} \frac{\boldsymbol{v}_j \boldsymbol{v}_j^\top}{d_j^2}, \qquad \mathbb{V}\mathrm{ar}(\boldsymbol{v}_j^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = \frac{\sigma^2}{d_j^2}.$$

- $d_j^2 = \|\boldsymbol{X}\boldsymbol{v}_j\|^2$ measures how much the data vary in the direction $\boldsymbol{v}_j$.
- A direction in which the predictors barely vary ($d_j \approx 0$) carries almost no information about $\boldsymbol{v}_j^\top \boldsymbol{\beta}$: its estimate has enormous variance.
- $\mathbb{E}\|\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{\beta}\|^2 = \sigma^2 \sum_j d_j^{-2}$ is dominated by the smallest d_j.

Two questions for today:
(i) what are these directions $\boldsymbol{v}_j$, statistically? (PCA).
(ii) What happens if we simply *discard* the poorly determined ones? (PCR).

PCA: the question

Pearson (1901); Hotelling (1933). Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ be a random vector with $\mathbb{E}\boldsymbol{X} = \boldsymbol{\mu}$ and $\mathbb{V}\mathrm{ar}(\boldsymbol{X}) = \boldsymbol{\Sigma}$.

Principal components. The *first* principal component is $Y_1 = \boldsymbol{a}_1^\top(\boldsymbol{X} - \boldsymbol{\mu})$, where

$$\boldsymbol{a}_1 = \underset{\|\boldsymbol{a}\|=1}{\arg\max}\, \mathbb{V}\mathrm{ar}(\boldsymbol{a}^\top \boldsymbol{X}) = \underset{\|\boldsymbol{a}\|=1}{\arg\max}\, \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}.$$

For $k \geq 2$, $Y_k = \boldsymbol{a}_k^\top(\boldsymbol{X} - \boldsymbol{\mu})$, where $\boldsymbol{a}_k$ maximizes $\boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}$ over all unit vectors $\boldsymbol{a}$ with

$$\mathbb{C}\mathrm{ov}(\boldsymbol{a}^\top \boldsymbol{X},\, Y_j) = \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}_j = 0, \qquad j = 1, \ldots, k-1.$$

- The constraint $\|\boldsymbol{a}\| = 1$ is essential: $\mathbb{V}\mathrm{ar}(c\, \boldsymbol{a}^\top \boldsymbol{X}) = c^2 \mathbb{V}\mathrm{ar}(\boldsymbol{a}^\top \boldsymbol{X})$ is unbounded in c.
- Each new component is the most variable linear combination that is *uncorrelated* with the earlier ones.

Two facts from linear algebra

Spectral theorem

A symmetric $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$ can be written $\boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top = \sum_{j=1}^{p} \lambda_j \boldsymbol{\gamma}_j \boldsymbol{\gamma}_j^\top$, with $\boldsymbol{\Gamma} = [\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_p]$ orthogonal and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p$. If $\boldsymbol{\Sigma} \succeq 0$, all $\lambda_j \geq 0$.

Rayleigh quotient bounds

For every unit vector $\boldsymbol{a}$: $\quad \lambda_p \leq \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a} \leq \lambda_1$.

Proof. Put $\boldsymbol{c} = \boldsymbol{\Gamma}^\top \boldsymbol{a}$; then $\|\boldsymbol{c}\| = \|\boldsymbol{a}\| = 1$ and

$$\boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a} = \boldsymbol{c}^\top \boldsymbol{\Lambda} \boldsymbol{c} = \sum_{j=1}^{p} \lambda_j c_j^2,$$

a weighted average of the λ_j with weights c_j^2 summing to one. Hence it lies in $[\lambda_p, \lambda_1]$, with the upper bound attained at $\boldsymbol{a} = \boldsymbol{\gamma}_1$ ($\boldsymbol{c} = \boldsymbol{e}_1$). $\square$

The Main Theorem

Theorem (principal components are eigenvectors)

Let $\boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\lambda_1 > \cdots > \lambda_p > 0$. Then $\boldsymbol{a}_k = \pm\boldsymbol{\gamma}_k$, and

$$Y_k = \boldsymbol{\gamma}_k^\top(\boldsymbol{X} - \boldsymbol{\mu}), \qquad \mathbb{V}\mathrm{ar}(Y_k) = \lambda_k, \qquad \mathbb{C}\mathrm{ov}(Y_j, Y_k) = 0 \ (j \neq k).$$

Proof. $k = 1$ is the Rayleigh bound. Suppose $\boldsymbol{a}_j = \boldsymbol{\gamma}_j$ for $j < k$, and write $\boldsymbol{a} = \sum_j c_j\boldsymbol{\gamma}_j$. The constraints read

$$0 = \boldsymbol{a}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}_j = \lambda_j c_j \quad \Longrightarrow \quad c_j = 0, \qquad j < k.$$

So $\boldsymbol{a}^\top\boldsymbol{\Sigma}\boldsymbol{a} = \sum_{j\geq k}\lambda_j c_j^2 \leq \lambda_k \sum_{j\geq k} c_j^2 = \lambda_k$, with equality iff $\boldsymbol{c} = \pm\boldsymbol{e}_k$, i.e. $\boldsymbol{a} = \pm\boldsymbol{\gamma}_k$. □

- In vector form: $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $\mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \boldsymbol{\Gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\Gamma} = \boldsymbol{\Lambda}$.
- PCA is a rotation of the coordinate system to the eigenvectors of $\boldsymbol{\Sigma}$; it uses $\boldsymbol{\Sigma}$ only.

Properties of principal components

With $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$:

1. **Total variance is preserved:** $\sum_j \mathbb{V}\mathrm{ar}(Y_j) = \sum_j \lambda_j = \operatorname{tr} \boldsymbol{\Sigma} = \sum_j \mathbb{V}\mathrm{ar}(X_j)$.
2. **Generalized variance is preserved:** $\det \mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \prod_j \lambda_j = \det \boldsymbol{\Sigma}$.
3. **Proportion explained** by the first k components: $\dfrac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_p}$.
4. **Reconstruction:** $\boldsymbol{X} - \boldsymbol{\mu} = \boldsymbol{\Gamma}\boldsymbol{Y} = \sum_{j=1}^p Y_j \boldsymbol{\gamma}_j$.
5. **Loadings as correlations:** $\mathbb{C}\mathrm{ov}(\boldsymbol{X}, Y_j) = \boldsymbol{\Sigma}\boldsymbol{\gamma}_j = \lambda_j \boldsymbol{\gamma}_j$, so

$$\mathbb{C}\mathrm{orr}(X_i, Y_j) = \frac{\gamma_{ij}\sqrt{\lambda_j}}{\sqrt{\sigma_{ii}}}, \qquad \sum_{j=1}^p \mathbb{C}\mathrm{orr}^2(X_i, Y_j) = \frac{\sum_j \lambda_j \gamma_{ij}^2}{\sigma_{ii}} = 1.$$

6. **Singular $\boldsymbol{\Sigma}$:** if $\lambda_{r+1} = \cdots = \lambda_p = 0$, then $\boldsymbol{\gamma}_j^\top(\boldsymbol{X} - \boldsymbol{\mu}) = 0$ almost surely for $j > r$: $\boldsymbol{X}$ lives in an r-dimensional affine subspace.

Geometry: ellipsoids and their axes

If $\boldsymbol{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the density is constant on the ellipsoids

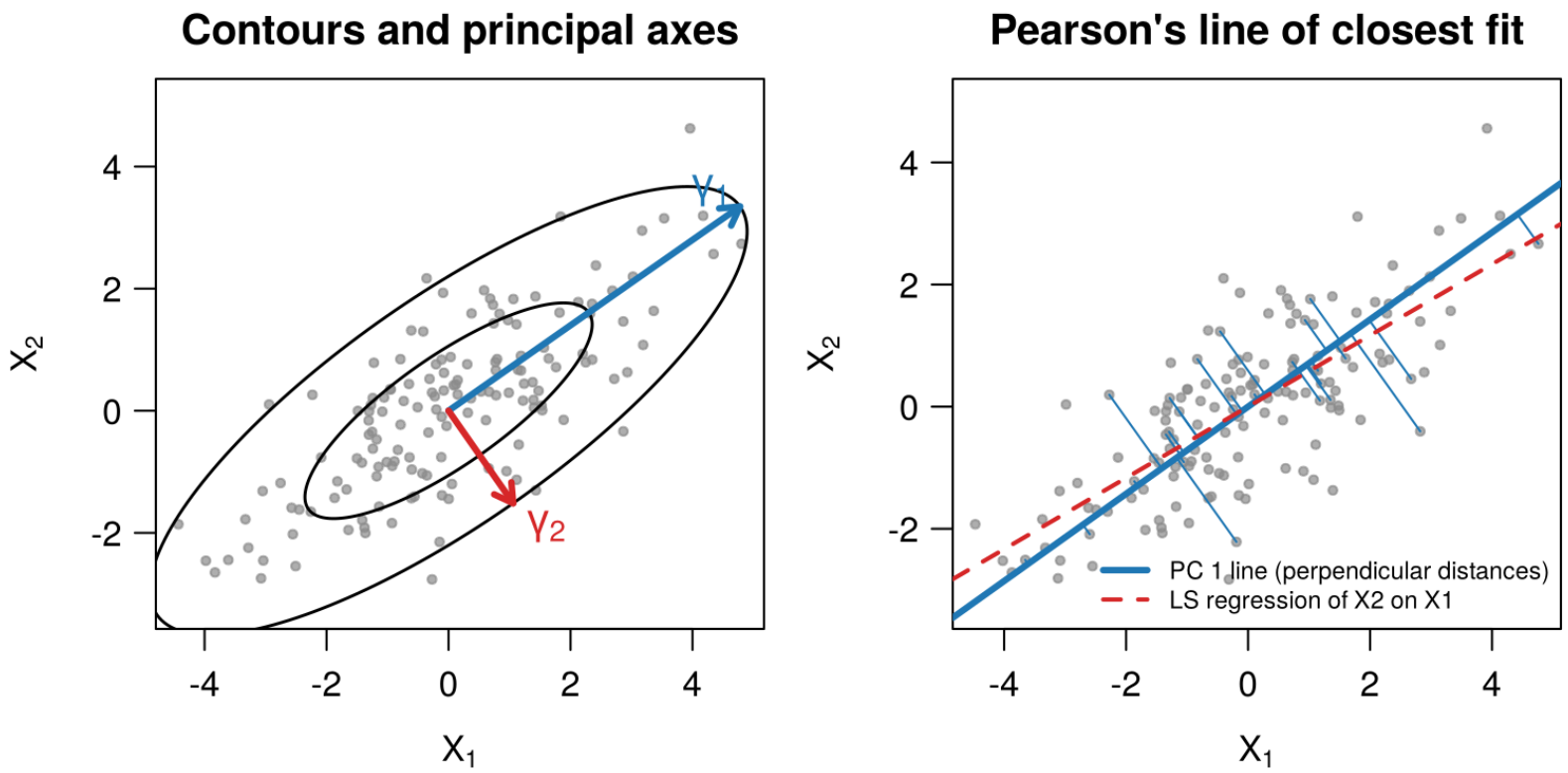
$$\mathcal{E}_c = \{\boldsymbol{x} : (\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) = c^2\}.$$

In principal-component coordinates $\boldsymbol{y} = \boldsymbol{\Gamma}^\top (\boldsymbol{x} - \boldsymbol{\mu})$,

$$(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) = \boldsymbol{y}^\top \boldsymbol{\Lambda}^{-1} \boldsymbol{y} = \sum_{j=1}^{p} \frac{y_j^2}{\lambda_j} = c^2.$$

- The axes of $\mathcal{E}_c$ point along $\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_p$, with half-lengths $c\sqrt{\lambda_1}, \ldots, c\sqrt{\lambda_p}$.
- PC 1 is the *longest* axis of the ellipsoid; the last PC is the *shortest*.

PCA in pictures



150 draws from $\mathcal{N}_2(\mathbf{0}, \mathbf{\Sigma})$, $\sigma_1 = 2$, $\sigma_2 = 1.5$, $\rho = 0.8$; $\lambda_1 = 5.68$, $\lambda_2 = 0.57$. **Left:** contours and principal axes. **Right:** the PC 1 line minimizes the sum of squared *perpendicular* distances (Pearson, 1901); least squares minimizes *vertical* distances.

Optimality: best low-dimensional approximation

Theorem (Ky Fan; optimal projection)

For every $p \times k$ matrix $\boldsymbol{B}$ with $\boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_k$,

$$\mathbb{E}\left\|(\boldsymbol{X} - \boldsymbol{\mu}) - \boldsymbol{B}\boldsymbol{B}^\top(\boldsymbol{X} - \boldsymbol{\mu})\right\|^2 = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B}) \;\geq\; \sum_{j>k} \lambda_j,$$

with equality when $\boldsymbol{B} = \boldsymbol{\Gamma}_k = [\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_k]$.

Proof sketch. $\boldsymbol{I} - \boldsymbol{B}\boldsymbol{B}^\top$ is a projection, so the left side is $\mathbb{E}\operatorname{tr}\{(\boldsymbol{I} - \boldsymbol{B}\boldsymbol{B}^\top)(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\} = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B})$. Ky Fan's maximum principle (Fan, 1949) says $\max_{\boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_k} \operatorname{tr}(\boldsymbol{B}^\top \boldsymbol{\Sigma} \boldsymbol{B}) = \lambda_1 + \cdots + \lambda_k$. □

- Two views of PCA agree: *maximize retained variance* (Hotelling) $\Longleftrightarrow$ *minimize reconstruction error* (Pearson).
- The k-dimensional subspace $\operatorname{span}(\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_k)$ is unique iff $\lambda_k > \lambda_{k+1}$.

Uncorrelated variables: PCA only permutes

Proposition

Suppose $X_1, \ldots, X_p$ are independent (it suffices: uncorrelated) with *distinct* variances $\sigma_1^2, \ldots, \sigma_p^2$. Let π order them, $\sigma^2_{\pi(1)} > \sigma^2_{\pi(2)} > \cdots > \sigma^2_{\pi(p)}$. Then

$$\gamma_k = \pm \boldsymbol{e}_{\pi(k)}, \qquad Y_k = \pm\big(X_{\pi(k)} - \mu_{\pi(k)}\big), \qquad \lambda_k = \sigma^2_{\pi(k)}.$$

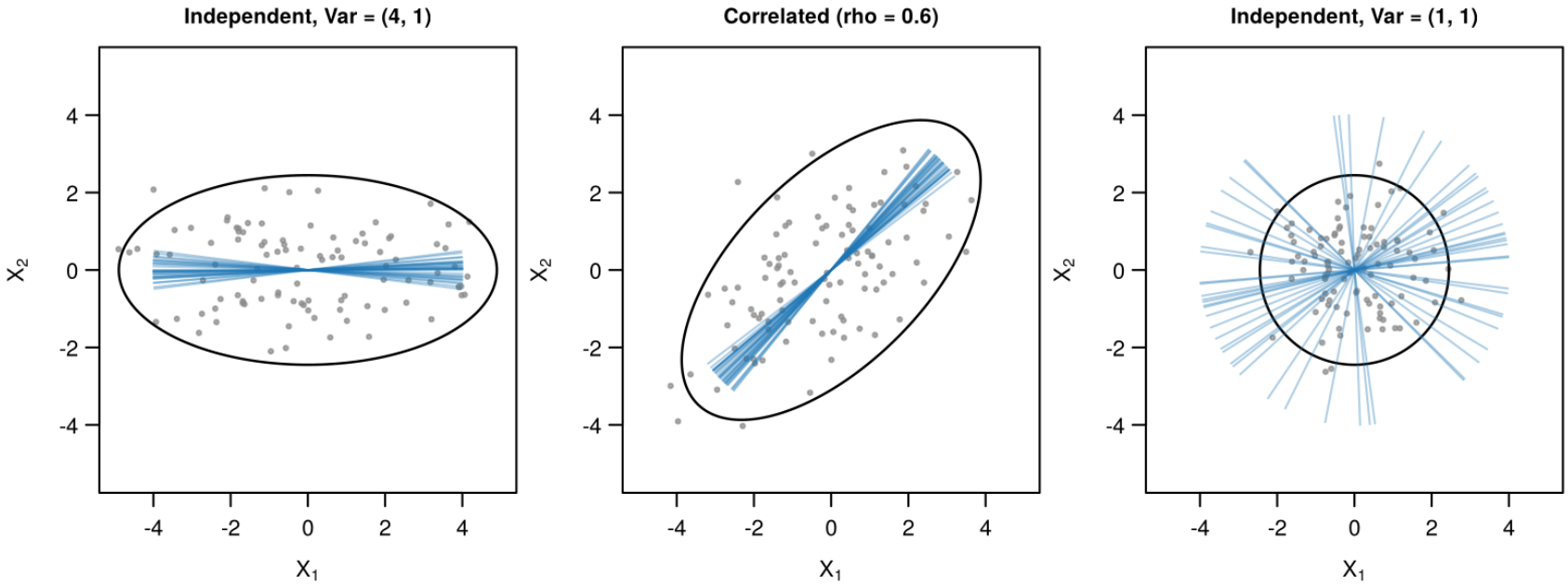
The principal components are the original (centred) variables, sorted by variance.

Proof. $\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_p^2)$, so $\boldsymbol{\Sigma}\boldsymbol{e}_i = \sigma_i^2 \boldsymbol{e}_i$: the coordinate vectors are eigenvectors. Distinct eigenvalues have one-dimensional eigenspaces, so these are the only unit eigenvectors (up to sign). Order them by eigenvalue. □

Uncorrelated variables: remarks

- **Converse.** If every γ_k is $\pm$ a coordinate vector, then $\mathbf{\Gamma}$ is a signed permutation matrix and $\mathbf{\Sigma} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}^\top$ is diagonal.
- PCA has nothing to offer when the variables are uncorrelated: it can only re-order them.
- Only $\mathbf{\Sigma}$ matters: independence is not needed, and non-normal data with diagonal $\mathbf{\Sigma}$ behave the same way.

Uncorrelated versus correlated: pictures



Each panel: one sample of $n = 100$ with its 95% population contour; blue lines are the *sample* PC 1 directions from 30 independent samples of the same population.

- Left: independent, variances $(4, 1)$: PC 1 $\approx X_1$, as the proposition says (spread of angles 3.6°).
- Middle: correlation 0.6: PC 1 is a rotated combination (3.7°).
- Right: independent, equal variances: population PCs are not unique, and sample PC 1 points anywhere (43°).

Sample PCA and the SVD

Data matrix $\boldsymbol{X}$ $(n \times p)$, column-centred. Sample covariance $\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X}/(n-1)$. With the singular value decomposition

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top, \qquad \boldsymbol{U}^\top \boldsymbol{U} = \boldsymbol{V}^\top \boldsymbol{V} = \boldsymbol{I}_p, \quad d_1 \geq \cdots \geq d_p \geq 0,$$

we have $\boldsymbol{S} = \boldsymbol{V} \frac{\boldsymbol{D}^2}{n-1} \boldsymbol{V}^\top$. Hence:

- sample PC directions = columns $\boldsymbol{v}_j$ of $\boldsymbol{V}$; sample eigenvalues $\ell_j = d_j^2/(n-1)$;
- sample PC scores $\boldsymbol{z}_j = \boldsymbol{X}\boldsymbol{v}_j = d_j \boldsymbol{u}_j$, with sample variance ℓ_j and zero sample correlation between different j.

Theorem (Eckart and Young, 1936)

Among all $n \times p$ matrices $\boldsymbol{A}$ of rank at most k,

$$\min_{\boldsymbol{A}} \|\boldsymbol{X} - \boldsymbol{A}\|_F^2 = \sum_{j>k} d_j^2, \qquad \text{attained at} \quad \boldsymbol{X}_k = \sum_{j \leq k} d_j \boldsymbol{u}_j \boldsymbol{v}_j^\top = \boldsymbol{Z}_k \boldsymbol{V}_k^\top.$$

Eckart and Young (1936): the sample analogue of the optimal-projection theorem: $\boldsymbol{X}_k$ projects every row onto $\mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$.

How many components to keep?

Some usual rules (Jolliffe, 2002; Johnson and Wichern, 2007):

1. **Cumulative proportion:** the smallest k with $\sum_{j \le k} \ell_j / \sum_j \ell_j \ge$ a threshold such as 80% or 90%.
2. **Average-eigenvalue rule:** keep components with $\ell_j > \bar{\ell} = \mathrm{tr}(\boldsymbol{S})/p$; for the correlation matrix, $\ell_j > 1$ (Kaiser, 1960).
3. **Scree plot:** plot ℓ_j against j and keep the components before the curve flattens (Cattell, 1966).

All three rules look only at $\boldsymbol{X}$. That is right for *describing* $\boldsymbol{X}$. For *predicting* a response it can be badly wrong, as we will see in PCR.

The regression model in PC coordinates

Model: $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{X}$ centred, $\operatorname{rank} \boldsymbol{X} = p$, $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$. Then $\widehat{\beta}_0 = \bar{y}$, and we may take $\boldsymbol{y}$ centred. Define

$$\boldsymbol{\gamma} = \boldsymbol{V}^\top \boldsymbol{\beta} \ \text{(coordinates of } \boldsymbol{\beta} \text{ along the PCs)}, \qquad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{D} \ \text{(scores)}.$$

Since $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{Z}\boldsymbol{\gamma}$, the model is $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, with orthogonal regressors $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{D}^2$.

OLS in these coordinates:

$$\widehat{\boldsymbol{\gamma}} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{D}^{-1} \boldsymbol{U}^\top \boldsymbol{y}, \qquad \widehat{\gamma}_j = \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j}, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}.$$

- The $\widehat{\gamma}_j$ are uncorrelated (independent under normality), $\mathbb{E}\widehat{\gamma}_j = \gamma_j$, $\mathbb{V}\mathrm{ar}(\widehat{\gamma}_j) = \sigma^2 / d_j^2$.
- $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}\widehat{\boldsymbol{\gamma}} = \sum_{j=1}^{p} \widehat{\gamma}_j \boldsymbol{v}_j$.

Principal components regression

Massy (1965); Jolliffe (2002), Ch. 8.

PCR with M components ($0 \leq M \leq p$): regress $\boldsymbol{y}$ on the first M score vectors $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_M$, and transform back:

$$\widehat{\boldsymbol{\beta}}_M = \sum_{j=1}^{M} \widehat{\gamma}_j \, \boldsymbol{v}_j = \sum_{j=1}^{M} \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j} \, \boldsymbol{v}_j = \boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}.$$

- Because the scores are orthogonal, $\widehat{\gamma}_j$ does not depend on which other components are in the model: PCR fits are *nested*.
- $\widehat{\boldsymbol{\beta}}_p = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$; $\widehat{\boldsymbol{\beta}}_0 = \mathbf{0}$ (intercept-only model).
- $\widehat{\boldsymbol{\beta}}_M$ is a coefficient vector for *all* p original predictors: PCR reduces dimension but does *not* select variables.
- The directions $\boldsymbol{v}_j$ and their order are determined by $\boldsymbol{X}$ alone; $\boldsymbol{y}$ enters only through $\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M$.

Three characterizations of PCR

Proposition

With $\boldsymbol{V}_{-M} = [\boldsymbol{v}_{M+1}, \ldots, \boldsymbol{v}_p]$ and $\boldsymbol{X}_M = \boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{V}_M^\top$ (the best rank-M approximation of $\boldsymbol{X}$):

(a) $\widehat{\boldsymbol{\beta}}_M = \boldsymbol{V}_M \boldsymbol{V}_M^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$ (project OLS onto the leading PC directions);

(b) $\widehat{\boldsymbol{\beta}}_M = \boldsymbol{X}_M^+ \boldsymbol{y}$ (OLS with $\boldsymbol{X}$ replaced by its rank-M approximation);

(c) $\widehat{\boldsymbol{\beta}}_M = \arg\min \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 : \ \boldsymbol{V}_{-M}^\top \boldsymbol{\beta} = \boldsymbol{0} \right\}$ (restricted least squares).

Proof. (a) $\boldsymbol{V}_M^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}_M^\top \boldsymbol{V} \widehat{\boldsymbol{\gamma}} = (\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M)^\top$. (b) The Moore–Penrose inverse of $\boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{V}_M^\top$ is $\boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top$. (c) The constraint means $\boldsymbol{\beta} = \boldsymbol{V}_M \boldsymbol{c}$; then $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{V}_M \boldsymbol{c}\|^2 = \|\boldsymbol{y} - \boldsymbol{U}_M \boldsymbol{D}_M \boldsymbol{c}\|^2$ is minimized at $\boldsymbol{c} = \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}$. □

By (c), PCR is OLS under $p - M$ *linear restrictions* $\boldsymbol{v}_j^\top \boldsymbol{\beta} = 0$, $j > M$, chosen by looking at $\boldsymbol{X}$ only. All of restricted-least-squares theory applies.

Fitted values and the hat matrix

Since $\boldsymbol{X}\boldsymbol{v}_j = d_j \boldsymbol{u}_j$,

$$\widehat{\boldsymbol{y}}_M = \bar{y}\mathbf{1} + \boldsymbol{X}\widehat{\beta}_M = \bar{y}\mathbf{1} + \sum_{j=1}^{M} \boldsymbol{u}_j \boldsymbol{u}_j^\top \boldsymbol{y} = \boldsymbol{H}_M \boldsymbol{y}, \qquad \boldsymbol{H}_M = \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top + \boldsymbol{U}_M \boldsymbol{U}_M^\top.$$

- $\mathbf{1} \perp \boldsymbol{u}_j$ (columns of $\boldsymbol{X}$ are centred), so $\boldsymbol{H}_M$ is symmetric, idempotent, of rank $M+1$: an orthogonal projection, and PCR is a linear smoother with $M+1$ degrees of freedom.
- Residual sum of squares and R^2 decompose additively over components:

$$\mathrm{RSS}_M = \|\boldsymbol{y} - \bar{y}\mathbf{1}\|^2 - \sum_{j \le M} (\boldsymbol{u}_j^\top \boldsymbol{y})^2, \qquad R^2_M = \sum_{j \le M} \frac{(\boldsymbol{u}_j^\top \boldsymbol{y})^2}{\|\boldsymbol{y} - \bar{y}\mathbf{1}\|^2}.$$

- Component j contributes $(\boldsymbol{u}_j^\top \boldsymbol{y})^2 = d_j^2 \widehat{\gamma}_j^2$ to the regression sum of squares. This does *not* have to decrease in j.

Bias and variance

From $\mathbb{E}\widehat{\gamma}_j = \gamma_j$ and $\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}$:

$$\mathbb{E}\widehat{\boldsymbol{\beta}}_M = \boldsymbol{V}_M \boldsymbol{V}_M^\top \boldsymbol{\beta}, \qquad \mathrm{bias}(\widehat{\boldsymbol{\beta}}_M) = -\sum_{j>M} \gamma_j \boldsymbol{v}_j, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_M) = \sigma^2 \sum_{j \le M} \frac{\boldsymbol{v}_j \boldsymbol{v}_j^\top}{d_j^2}.$$

Proposition (PCR reduces variance)

$$\mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}) - \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\beta}}_M) = \sigma^2 \sum_{j>M} d_j^{-2} \boldsymbol{v}_j \boldsymbol{v}_j^\top \succeq 0.$$

Hence $\mathbb{V}\mathrm{ar}(\boldsymbol{c}^\top \widehat{\boldsymbol{\beta}}_M) \le \mathbb{V}\mathrm{ar}(\boldsymbol{c}^\top \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}})$ for every $\boldsymbol{c} \in \mathbb{R}^p$.

- $\widehat{\boldsymbol{\beta}}_M$ is unbiased iff $\gamma_j = 0$ for all $j > M$, i.e. iff $\boldsymbol{\beta} \in \mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_M)$.
- The variance saved, $\sigma^2 \sum_{j>M} d_j^{-2}$ in total, is largest exactly when the discarded d_j are small: the collinear directions.
- Gauss–Markov is not violated: $\widehat{\boldsymbol{\beta}}_M$ is generally not *unbiased* (unless the restrictions hold).

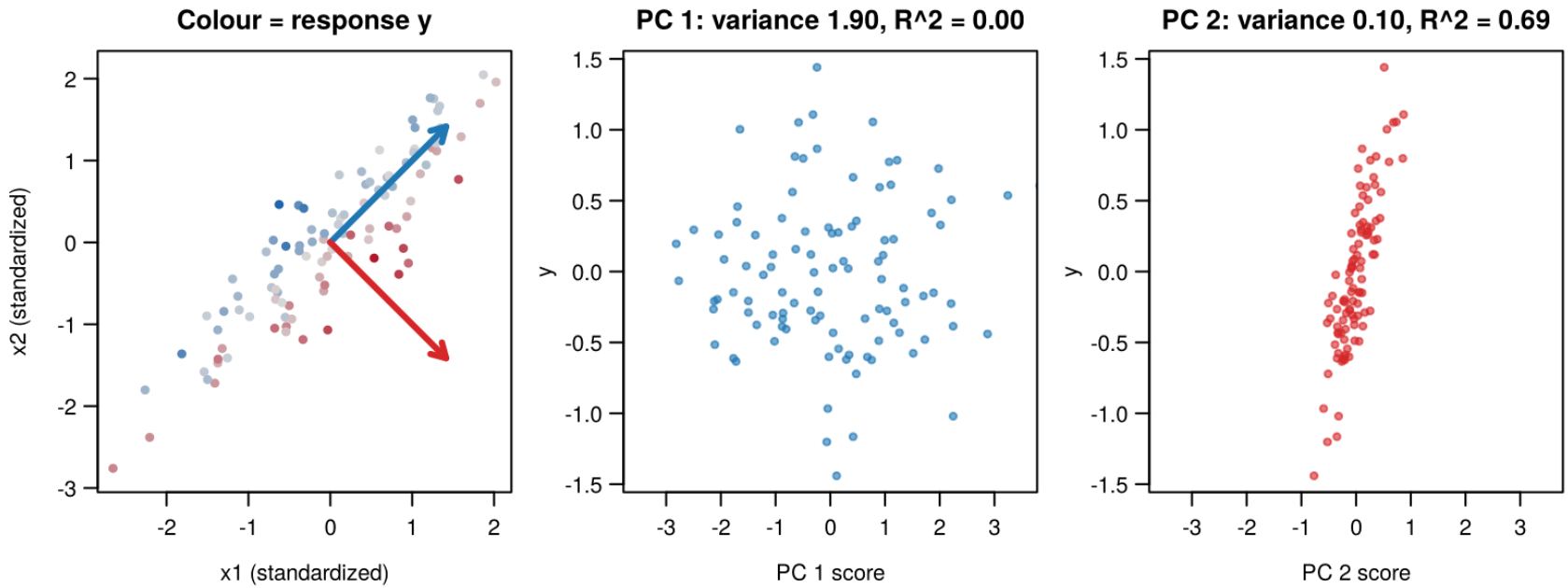
Which components matter for Y?

Population version: $Y = \beta_0 + \boldsymbol{\beta}^\top \boldsymbol{X} + \varepsilon$, $\mathbb{V}\mathrm{ar}(\boldsymbol{X}) = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$, $Y_j = \boldsymbol{\gamma}_j^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $c_j = \boldsymbol{\gamma}_j^\top \boldsymbol{\beta}$. Then

$$\mathbb{V}\mathrm{ar}(Y) = \sum_{j=1}^{p} \lambda_j c_j^2 + \sigma^2, \qquad \mathbb{C}\mathrm{ov}(Y, Y_j) = \lambda_j c_j, \qquad \mathbb{C}\mathrm{orr}^2(Y, Y_j) = \frac{\lambda_j c_j^2}{\mathbb{V}\mathrm{ar}(Y)}.$$

- The contribution of component j is $\lambda_j c_j^2$: a large variance λ_j helps, but c_j can be anything.
- **Nothing forces the response to depend on the high-variance components.** Jolliffe (1982) collects real examples where low-variance components were the predictive ones; Hadi and Ling (1998) give an example in which the last component carries essentially all of the fit.
- PCA ranks directions by λ_j; regression needs them ranked by $\lambda_j c_j^2$.

The weak spot in a picture



x_1, x_2 with correlation 0.9, $n = 100$, $y = x_1 - x_2 +$ noise. The response varies along the *short* axis of the cloud. PC 1 (variance 1.90) explains none of y; PC 2 (variance 0.10) explains 69%. PCR with $M = 1$ discards exactly the direction that matters.

Orthogonal predictors: PCR selects by spread

Recall: uncorrelated variables $\Rightarrow$ the PCs are the variables themselves, sorted by variance. The sample analogue:

Proposition

If $\boldsymbol{X}^\top \boldsymbol{X} = \mathrm{diag}(s_1^2, \ldots, s_p^2)$ with distinct s_j, then $\boldsymbol{v}_j = \pm \boldsymbol{e}_{\pi(j)}$ where $s_{\pi(1)} > \cdots > s_{\pi(p)}$, and

$$\widehat{\boldsymbol{\beta}}_M = \text{OLS on the } M \text{ predictors with the largest } s_j, \text{ zero elsewhere.}$$

- PCR then becomes *variable selection by the variance of the predictors*, a criterion that never looks at $\boldsymbol{y}$.
- It depends on units: rescaling one column changes s_j and can change which variables are kept (unlike OLS, whose fit is invariant to rescaling columns).
- If all s_j are equal (e.g. standardized orthogonal predictors), the PCs are not unique and PCR with $M < p$ is *not even well defined*. Ridge, by contrast, shrinks every coordinate by the same factor $s^2/(s^2 + \lambda)$.

Take-home messages

- **PCA** rotates to the eigenvectors of $\boldsymbol{\Sigma}$: $\boldsymbol{Y} = \boldsymbol{\Gamma}^\top(\boldsymbol{X} - \boldsymbol{\mu})$, $\mathbb{V}\mathrm{ar}(\boldsymbol{Y}) = \boldsymbol{\Lambda}$. PC k maximizes variance among directions uncorrelated with PCs $1, \ldots, k-1$; the leading k PCs give the best k-dimensional approximation (Ky Fan; Eckart–Young).
- Uncorrelated variables: the PCs are the variables, sorted by variance; with ties they are not unique. PCA is not scale invariant.
- Sample eigenvectors are unstable when eigenvalues are close: $\mathbb{V}\mathrm{ar} \propto \lambda_j\lambda_k/(\lambda_j - \lambda_k)^2$.
- **PCR** is restricted least squares with restrictions $\boldsymbol{v}_j^\top \boldsymbol{\beta} = 0$, $j > M$, and is a projection smoother with $M + 1$ degrees of freedom.
- MSE $= \sum_{j>M} \gamma_j^2 + \sigma^2 \sum_{j \le M} d_j^{-2}$. PCR beats OLS for every linear combination iff $\sum_{j>M} d_j^2 \gamma_j^2 \le \sigma^2$.
- PCA ranks directions by λ_j; prediction needs $\lambda_j c_j^2$. Choose M by C_p or cross-validation, never by $\boldsymbol{X}$-variance alone.

Practice problems

1. Prove the Rayleigh bounds and the main theorem (PC k is $\boldsymbol{\gamma}_k$) when $\lambda_1 > \cdots > \lambda_p$. What changes if $\lambda_k = \lambda_{k+1}$?
2. Show that $\sum_j \mathbb{C}\mathrm{orr}^2(X_i, Y_j) = 1$ for every i.
3. Show that the principal components are a (signed) permutation of the centred variables if and only if $\boldsymbol{\Sigma}$ is diagonal with distinct diagonal entries.
4. For $\boldsymbol{\Sigma} = (1-\rho)\boldsymbol{I}_p + \rho \mathbf{1}\mathbf{1}^\top$ $(0 < \rho < 1)$, find all eigenvalues and principal components. Which are unique?
5. Derive the eigenvalues and the angle formula $\tan 2\theta = 2\rho\sigma_1\sigma_2/(\sigma_1^2 - \sigma_2^2)$ in the bivariate case.
6. Prove the three characterizations of PCR, and show that $\boldsymbol{H}_M$ is an orthogonal projection of rank $M+1$.
7. Prove the lemma: for $\boldsymbol{A} \succeq 0$ and $\boldsymbol{b} \in \mathrm{col}(\boldsymbol{A})$, $\boldsymbol{A} - \boldsymbol{b}\boldsymbol{b}^\top \succeq 0 \iff \boldsymbol{b}^\top \boldsymbol{A}^+ \boldsymbol{b} \leq 1$. Deduce the MSE-matrix theorem.
8. Show that ridge with any $\lambda > 0$ never sets a coordinate γ_j to zero, and compute $\lim_{\lambda\to\infty}$ and $\lim_{\lambda\to 0}$ of $\widehat{\boldsymbol{\beta}}_\lambda$.
9. Prove the leave-one-out formula for PCR (hint: Sherman–Morrison, or work with the projection $\boldsymbol{H}_M$).

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